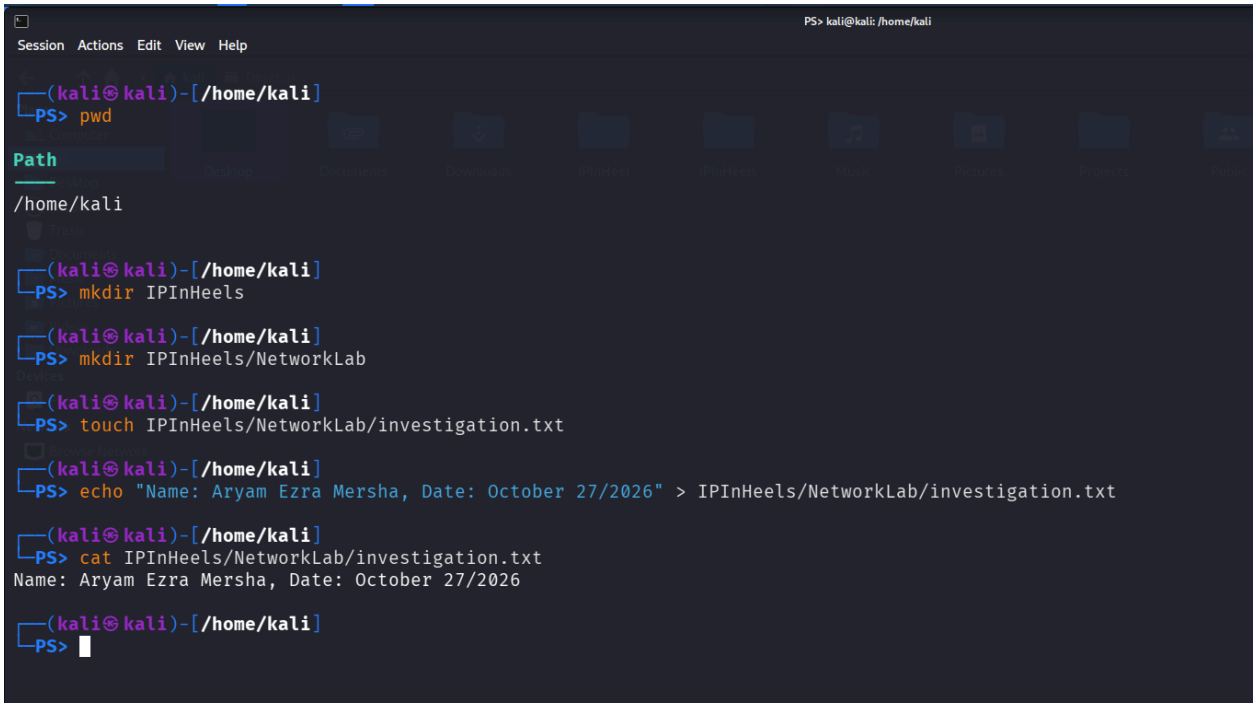
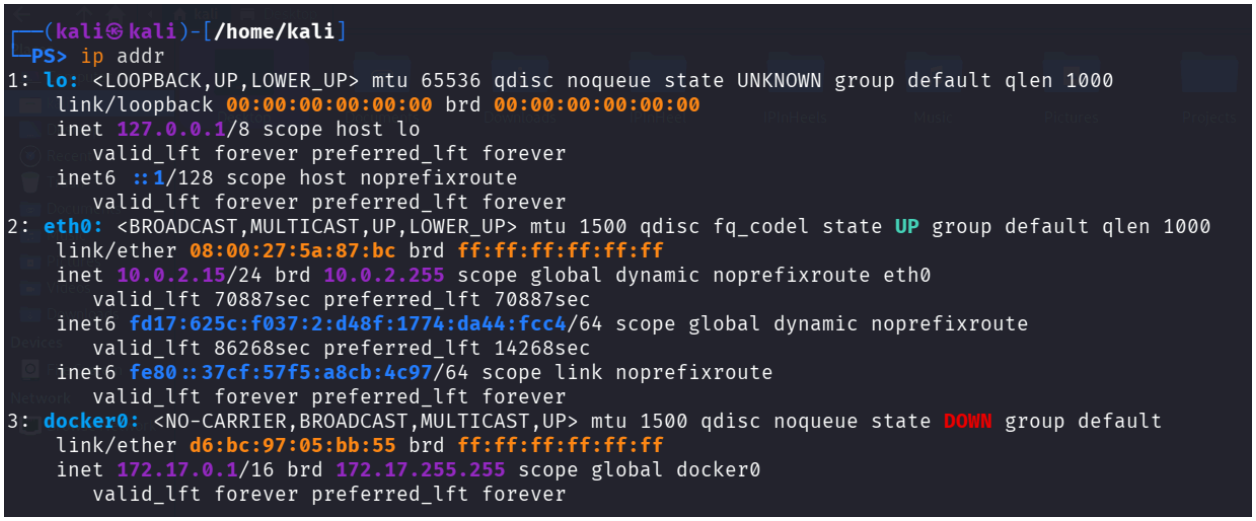
Task 1 : Navigate the Linux terminal



Task 2 : Investigate a network connection



```
┌──(kali㉿kali)-[/home/kali]
└─PS> ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown

┌──(kali㉿kali)-[/home/kali]
└─PS> ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=56.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=67.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=53.5 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=70.9 ms

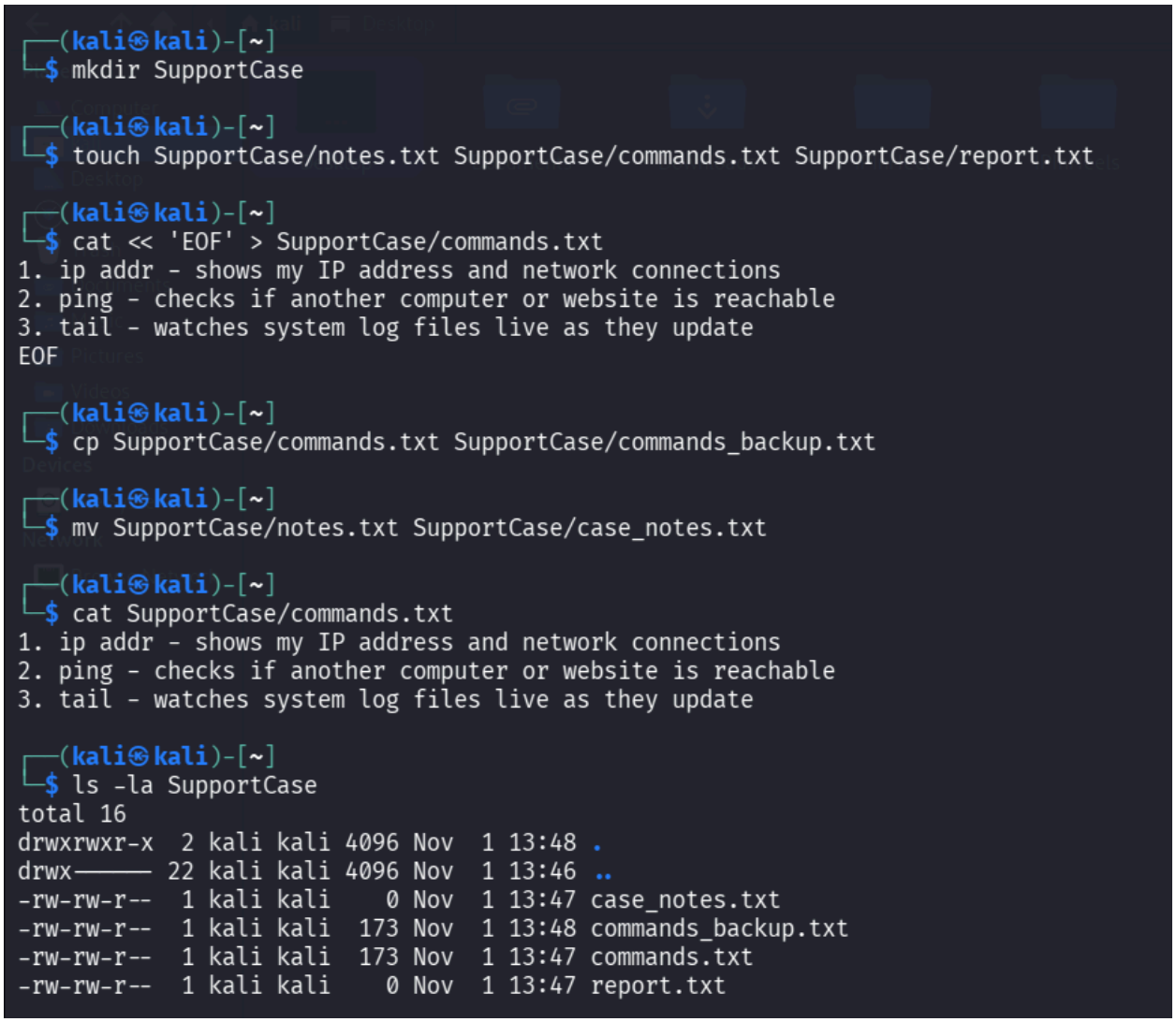
── 8.8.8.8 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 53.545/61.922/70.899/7.286 ms
```

```
┌──(kali㉿kali)-[/home/kali]
└─PS> traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  0.611 ms  0.555 ms  0.534 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

The `ip addr` command displayed active network interfaces including the loopback (`lo`) and VirtualBox NAT interface (`eth0`) configured with IP `10.0.2.15`. The `ip route` command verified the default gateway (`10.0.2.2`). Successful `ping` execution confirmed stable external reachability with zero packet loss, while `traceroute` revealed the initial gateway hop before hitting public ICMP filtering rules.

One possible reason why a ping or traceroute might fail is network firewalls or routers often drop or block ICMP packets (such as echo requests and TTL time-exceeded messages) for security or policy enforcement reasons, preventing devices from mapping internal network infrastructure even when the destination host is completely online.

Task 3: Linux File Management Challenge



Task 4: Build a Personal Tech Skills Portfolio

```
┌──(kali㉿kali)-[~]
└─$ cat << 'EOF' > SupportCase/README.md
My First Linux Support Lab

Overview
This repository documents practical foundational skills acquired in Linux terminal navigation, file management, and basic network troubleshooting.

Key Commands Learned and When to Use Them
1. ip addr: Used during network diagnostics to inspect interface statuses, MAC addresses, and assigned IP configurations.
2. ping: Utilized for quick reachability verification and checking round-trip latency to remote gateways or servers.
3. cp and mv: Essential file management utilities used for backing up configuration scripts and structuring support case documentation.

Troubleshooting Log and Problem Solved
Problem: Initially experienced syntax and redirection errors when attempting to use multi-line text input blocks inside a Windows PowerShell prompt rather than the Linux shell.
Solution: Switched to the correct Linux shell environment and verified directory structures to successfully write and manage support case files.

7-Day Technical Skill Goal
During the next seven days, I plan to practice analyzing packet captures using network diagnostic tools like wireshark and packet filters to deepen my understanding of network traffic behav
ior.
EOF
```

```
┌──(kali㉿kali)-[~]
└─$ cat SupportCase/README.md
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ior.

┌──(kali㉿kali)-[~]
└─$
```

https://github.com/AryamEzra/IP-In-Heels